

Notes on the b-Calculus and the APS Index Theorem

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1 Elliptic Fredholm theory on closed manifolds

The b-calculus is a boundary refinement of ordinary elliptic theory, so we first record the closed-manifold argument that will later fail at the cylindrical end. Let M be closed and let

$$P : C^\infty(M; E) \longrightarrow C^\infty(M; F)$$

be a differential or classical pseudodifferential operator of order m . Its principal symbol

$$\sigma_m(P)(x, \xi) : E_x \longrightarrow F_x$$

is obtained by retaining the terms homogeneous of degree m in ξ . The operator is elliptic when this map is invertible for every $\xi \neq 0$.

For every $s \in \mathbb{R}$, an operator of order m extends boundedly as

$$P : H^s(M; E) \longrightarrow H^{s-m}(M; F).$$

Ellipticity permits the inverse symbol to be quantized and corrected recursively.

Theorem 1.1 (Elliptic parametrix). *If P is elliptic, there is $Q \in \Psi^{-m}(M; F, E)$ such that*

$$QP = I - R_L, \quad PQ = I - R_R, \quad R_L, R_R \in \Psi^{-\infty}(M).$$

Proof. Let $p \sim p_m + p_{m-1} + \dots$ be the complete symbol of P . Choose $q_{-m} = p_m^{-1}$ away from the zero section. If $q_{-m}, \dots, q_{-m-j+1}$ are known, the degree $-j$ part of $q\#p - I$ is affine in the unknown q_{-m-j} , with coefficient p_m . Multiplication by p_m^{-1} therefore determines q_{-m-j} and removes that term. Borel summation produces a classical symbol q for which $q\#p - I$ has order $-\infty$. Repeating on the other side and applying a smoothing correction gives one Q with both displayed identities. $\square$

The logic of the construction is:

$$\begin{array}{ccccc} P & \xrightarrow{\sigma_m} & p_m & \xrightarrow{\text{invert}} & p_m^{-1} \\ & & & & \downarrow \text{symbol recursion} \\ \text{Fredholm} & \xleftarrow{\text{Atkinson}} & I - \text{compact} & \xleftarrow{\text{compose}} & Q \end{array}$$

Theorem 1.2 (Closed elliptic Fredholm theorem). *For every s , the map*

$$P : H^s(M; E) \longrightarrow H^{s-m}(M; F)$$

is Fredholm. Its kernel is smooth, its range is closed, and

$$\text{coker } P \cong \ker P^*.$$

If $Pu \in H^{r-m}$, then $u \in H^r$.

Proof. The parametrix gives

$$u = QPu + R_L u.$$

This proves elliptic regularity because Q gains m derivatives and R_L is smoothing. On a closed manifold, a smoothing operator factors through $H^{s+N} \hookrightarrow H^s$, which is compact by Rellich's lemma. Thus P is invertible modulo compact operators and is Fredholm by Atkinson's theorem. The orthogonal complement of its closed range is $\ker P^*$, giving the cokernel statement. $\square$

The analytic index

$$\text{ind } P = \dim \ker P - \dim \ker P^*$$

is independent of s and constant in continuous families of elliptic operators. A lower-order perturbation preserves the principal symbol and hence the index.

Lemma 1.3 (Fedosov trace formula). *Suppose Q is a parametrix for the Fredholm operator P and*

$$S_L = I - QP, \quad S_R = I - PQ$$

are trace class. Then

$$\text{ind } P = \text{Tr}(S_L) - \text{Tr}(S_R).$$

Proof. Let G be the generalized inverse of P , equal to the inverse from $(\ker P)^\perp$ onto $\text{ran } P$ and zero on $\ker P^*$. If Π_0, Π_1 are the orthogonal projections onto $\ker P, \ker P^*$, then

$$GP = I - \Pi_0, \quad PG = I - \Pi_1.$$

Hence $\text{ind } P = \text{Tr}(\Pi_0) - \text{Tr}(\Pi_1)$. Subtracting these identities from the identities for Q shows that $(Q - G)P$ and $P(Q - G)$ are trace class. Cyclicity gives them equal traces, so

$$\text{Tr}(S_L) - \text{Tr}(S_R) = \text{Tr}(\Pi_0) - \text{Tr}(\Pi_1).$$

$\square$

Discussion. Both hypotheses in the closed theorem matter. Ellipticity produces smoothing remainders; compactness of the base turns the Sobolev gain into a compact map. On a cylinder, translations of a fixed bump remain bounded but have no convergent subsequence. The b-calculus must therefore control not only high frequency, through the principal symbol, but also escape to infinity, through the normal operator.

2 From the closed index theorem to the APS formula

Let X be a compact even-dimensional Riemannian manifold with boundary Y . Assume that the metric and all bundle data are of product type on a collar $[0, 1)_u \times Y$. Let $E = E^+ \oplus E^-$ be a graded Hermitian Clifford module and let

$$D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}$$

be its Dirac operator. On the collar we use the convention

$$D = c(du)(\partial_u + A),$$

where A is the self-adjoint tangential Dirac operator and $c(du)A = -Ac(du)$.

2.1 Closed manifolds

When $\partial X = \emptyset$, the heat operators are smoothing and trace class. Define

$$Q_t = D^- \int_0^t e^{-sD^+D^-} ds.$$

Functional calculus and the intertwining relation $D^- e^{-sD^+D^-} = e^{-sD^-D^+} D^-$ give

$$D^+ Q_t = I - e^{-tD^+D^-}, \quad Q_t D^+ = I - e^{-tD^-D^+}.$$

Fedosov's trace formula therefore yields the McKean–Singer identity

$$\text{ind } D^+ = \text{Tr}(e^{-tD^-D^+}) - \text{Tr}(e^{-tD^+D^-}) = \text{Str}(e^{-tD^2})$$

for every $t > 0$. The local heat expansion then gives

$$\text{ind } D^+ = \int_X \left[\hat{A}(TX, \nabla^{TX}) \text{ch}(E/S, \nabla^{E/S}) \right]_{[n]}.$$

2.2 The cylindrical end

Attach $(-\infty, 0] \times Y$ to X and extend D by its product formula. The resulting complete manifold is denoted X_∞ . The operator

$$D^+ : H^1(X_\infty, E^+) \longrightarrow L^2(X_\infty, E^-)$$

is Fredholm precisely when A is invertible. Expand a section in an eigenbasis of A . On the mode with eigenvalue λ , the operator on the end reduces to $\partial_x + \lambda$. A spectral gap gives uniform exponential estimates. If $0 \in \text{spec}(A)$, translates of a fixed cutoff in a zero mode form a Weyl sequence at zero.

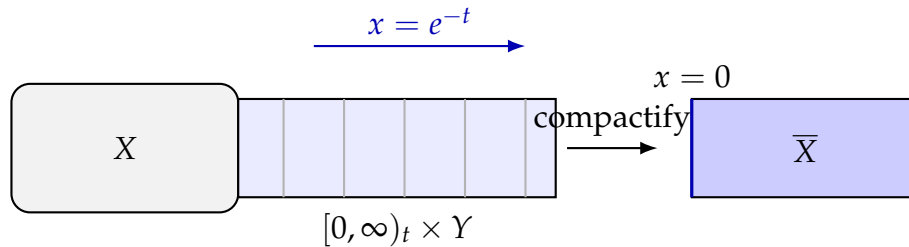


Figure 1: The infinite cylinder becomes a boundary hypersurface under $x = e^{-t}$. Translation in t becomes dilation in x .

Although D^+ is Fredholm when A is invertible, the two heat operators are not separately trace class: their kernels are translation invariant to leading order on the infinite cylinder. Their graded difference is handled by the b -trace introduced below.

For $\Re s$ large, define the eta function

$$\eta_A(s) = \sum_{\lambda \in \text{spec}(A) \setminus \{0\}} \text{sgn}(\lambda) |\lambda|^{-s}.$$

Its Mellin representation is

$$\eta_A(s) = \frac{1}{\Gamma\left(\frac{s+1}{2}\right)} \int_0^\infty t^{(s-1)/2} \operatorname{Tr}(Ae^{-tA^2}) dt.$$

The right side extends meromorphically and is regular at $s = 0$. With $h = \dim \ker A$, the Atiyah–Patodi–Singer formula is

$$\operatorname{ind} D_{\text{APS}}^+ = \int_X \left[\widehat{A}(TX) \operatorname{ch}(E/S) \right]_{[n]} - \frac{\eta_A(0) + h}{2}.$$

When A is invertible, $h = 0$, and this equals the L^2 index on X_∞ .

Question 2.1. *Why does attaching a cylinder help if the resulting manifold is noncompact?*

Answer. The cylinder does not remove the analytic difficulty; it puts that difficulty into a translation-invariant form. On the compact manifold with boundary, the APS condition is the nonlocal projection determined by the spectrum of A . On the cylinder, the same condition is encoded by square integrability. Indeed, on an A -eigenvector with eigenvalue λ , the equation $(\partial_x + A)u = 0$ becomes an elementary ODE, and exactly one spectral half gives decaying solutions on the outward end. Thus the APS projection and the L^2 condition select the same boundary data. The remaining translation-invariant part is what the normal operator and the b -trace measure. $\square$

3 The APS boundary problem

Let

$$\gamma : H^1(X; E^+) \longrightarrow H^{1/2}(Y; E^+|_Y)$$

be the trace map. Since A is self-adjoint and elliptic on the closed manifold Y , its spectral projections are order-zero pseudodifferential operators. Define

$$\Pi_{\geq 0}(A) = \sum_{\lambda \geq 0} \Pi_\lambda$$

and the APS domain

$$\mathcal{D}_{\text{APS}}(D^+) = \left\{ u \in H^1(X; E^+) : \Pi_{\geq 0}(A) \gamma u = 0 \right\}.$$

Thus the boundary value is allowed to lie only in the negative spectral subspace of A .

Green's formula fixes the complementary adjoint condition. With the collar convention above,

$$\langle D^+ u, v \rangle_X - \langle u, D^- v \rangle_X = -\langle c(du) \gamma u, \gamma v \rangle_Y.$$

Since $c(du)$ interchanges the positive and negative spectral subspaces, the adjoint of D_{APS}^+ is D^- with the complementary strict projection. This distinction at zero is responsible for the $\dim \ker A$ term in the index formula.

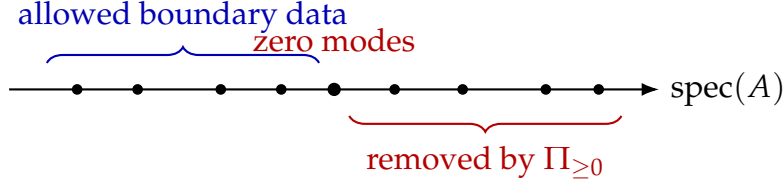


Figure 2: The APS condition is nonlocal: it uses the global spectral decomposition of the tangential operator rather than pointwise boundary values.

Theorem 3.1 (Fredholmness of the APS realization). *The operator*

$$D_{\text{APS}}^+ : \mathcal{D}_{\text{APS}}(D^+) \longrightarrow L^2(X; E^-)$$

is Fredholm. If A is invertible, its kernel and cokernel agree with those of the L^2 realization on X_∞ ; in particular,

$$\text{ind } D_{\text{APS}}^+ = \text{ind}_{L^2} D_{X_\infty}^+.$$

Proof. On the collar, expand

$$u(u, y) = \sum_{\lambda} u_{\lambda}(u) \phi_{\lambda}(y), \quad A \phi_{\lambda} = \lambda \phi_{\lambda}.$$

The principal part of D^+ is the direct sum of the one-dimensional operators $\partial_u + \lambda$. The APS condition sets $u_{\lambda}(0) = 0$ for $\lambda \geq 0$. Integration by parts in each mode gives the uniform collar estimate

$$\|u\|_{H^1([0, \varepsilon] \times Y)} \leq C \left(\|D^+ u\|_{L^2} + \|u\|_{L^2} + \|u\|_{H^1(\{u > \varepsilon/2\})} \right).$$

An interior elliptic estimate absorbs the last term and yields

$$\|u\|_{H^1(X)} \leq C (\|D^+ u\|_{L^2(X)} + \|u\|_{L^2(X)})$$

on the APS domain. The same estimate holds for the adjoint boundary problem. Since $H^1(X) \hookrightarrow L^2(X)$ is compact, the standard compactness argument gives finite-dimensional kernels and closed range; the adjoint estimate gives finite-dimensional cokernel. Hence the realization is Fredholm.

Assume now that A is invertible and write the outward cylinder coordinate as $t = -u \geq 0$. A boundary eigenmode extends as a solution of the product equation by

$$u_{\lambda}(t) = e^{\lambda t} u_{\lambda}(0).$$

This extension is square integrable exactly when $\lambda < 0$, precisely the modes allowed by the APS condition. Restriction and extension therefore identify the kernels. Applying the same argument to the adjoint identifies the cokernels, proving equality of indices. $\square$

Question 3.2. *Why is an elliptic differential operator on a manifold with boundary not automatically Fredholm?*

Answer. The interior principal symbol controls high-frequency behavior away from the boundary but does not select boundary data. For example, the $\bar{\partial}$ -operator on a disk has the infinite-dimensional space of holomorphic functions in its kernel if no boundary condition is imposed. An elliptic boundary condition removes half of the Cauchy data. The APS condition makes that choice spectrally; the b -calculus makes the same choice through decay on the attached cylinder. $\square$

4 Regularized b -integrals and b -traces

Put $\rho = e^u$ on the cylindrical end $u \in (-\infty, 0]$. Then ρ is a boundary defining function for the compactification and $dx = d\rho/\rho$. Suppose that a smooth function has an expansion

$$f(\rho, y) = f_0(y) + O(\rho^\epsilon), \quad \epsilon > 0,$$

with the same estimate after applying any number of b -derivatives. For $\Re z > 0$, set

$$F_f(z) = \int_X \rho^z f d\mu_b.$$

The contribution of f_0 is

$$\int_0^1 \rho^{z-1} d\rho \int_Y f_0 d\mu_Y = \frac{1}{z} \int_Y f_0 d\mu_Y.$$

The remainder is holomorphic for $\Re z > -\epsilon$. Hence

$$F_f(z) = \frac{1}{z} \int_Y f_0 d\mu_Y + G_f(z),$$

where G_f is holomorphic near zero.

Definition 4.1. The b -integral of f is the finite part

$${}^b \int_X f d\mu_b := \text{FP}_{z=0} F_f(z) = G_f(0).$$

The definition depends on the chosen boundary defining function, but the trace-defect formula below changes compatibly, and the final index is invariant.

Let K be a smoothing b -operator whose diagonal has an expansion of the preceding form.

Definition 4.2. The b -trace is

$${}^b \text{Tr}(K) := {}^b \int_X \text{tr}_E K(p, p) d\mu_b(p).$$

For a graded bundle, the analogous definition uses the fibrewise supertrace.

Unlike the ordinary trace, the b -trace need not vanish on commutators. Its defect is a boundary term determined by the indicial family. The precise formula and its application to the heat kernel appear below.

Discussion. This finite part is not an arbitrary subtraction. If the cylinder is cut at $u = -T$, then the diagonal integral has the form $cT + C + o(1)$. The b -integral is the constant term C . The coefficient c depends only on the translation-invariant kernel at infinity, so the same boundary model that obstructs trace class also controls the trace defect.

5 Real blow-up and the b-double space

A smooth n -manifold with corners is locally modelled on

$$\mathbb{R}^{n,k} = [0, \infty)^k \times \mathbb{R}^{n-k}, \quad 0 \leq k \leq n,$$

with smooth transition maps in the usual extension sense. A boundary hypersurface is a connected codimension-one boundary face. At a point p , the codimension is the number of independent boundary defining functions that vanish at p .

Let $S \subset X$ be an embedded boundary face. Its real blow-up $[X; S]$ replaces S by the inward-pointing spherical normal bundle S^+NS . The blow-down map

$$\beta : [X; S] \longrightarrow X$$

is a diffeomorphism away from the new boundary hypersurface and collapses that hypersurface onto S .

For a compact manifold X with boundary, the b -double space is

$$X_b^2 = [X^2; \partial X \times \partial X].$$

Its boundary hypersurfaces are the front face ff and the lifts lb and rb of $\partial X \times X$ and $X \times \partial X$. The lifted diagonal Δ_b is the closure of the interior diagonal.

The geometry is summarized by the blow-down square

$$\begin{array}{ccc} \Delta_b & \hookrightarrow & X_b^2 \\ \beta|_{\Delta_b} \downarrow & & \downarrow \beta \\ \Delta & \hookrightarrow & X^2. \end{array} \quad \beta(\text{ff}) = \partial X \times \partial X.$$

The blow-down is an isomorphism away from $\partial X \times \partial X$. Over that corner, the front face records the ratio of the two normal variables and therefore remembers how the corner is approached.

Near ∂X , choose product coordinates (x, y) and (x', y') on the two factors. Two useful projective coordinate systems are

$$(x, s = x'/x, y, y') \quad \text{and} \quad (s' = x/x', x', y, y').$$

In the first chart, $\text{ff} = \{x = 0\}$ and $\text{rb} = \{s = 0\}$; in the second, $\text{ff} = \{x' = 0\}$ and $\text{lb} = \{s' = 0\}$. Near the interior of the front face one may instead use

$$r = x, \quad z = \log(x/x'), \quad y, \quad Y = y - y',$$

so that $\Delta_b = \{z = 0, Y = 0\}$.

Question 5.1. *Why not work directly on X^2 ?*

Answer. The ratio x/x' has no limit at $(x, x') = (0, 0)$ on X^2 . Kernels of translation-invariant operators on the cylinder depend on $\log(x/x')$, so they cannot be smooth functions of x, x' at that corner. Blow-up replaces the corner by the set of limiting ratios. In the projective chart $x' = sx$, the same kernel is a function of $-\log s$, which is smooth in the interior of the front face and has controlled asymptotics as $s \rightarrow 0$ or $s \rightarrow \infty$. The singularity has not been discarded; it has been separated into geometrically distinct boundary faces. $\square$

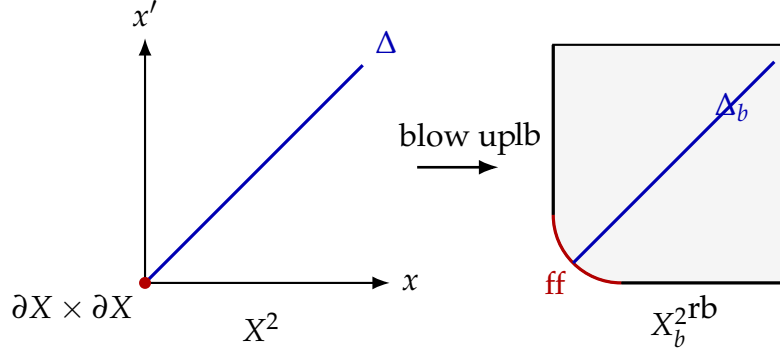


Figure 3: Blowing up the boundary corner replaces one point of approach by the front face of limiting normal ratios; the diagonal lifts transversely.

5.1 A one-dimensional model

On the full cylinder $\mathbb{R}_t$, consider

$$P = -\partial_t^2 + \epsilon^2, \quad \epsilon > 0.$$

Near the end $t = -\infty$, use the compactifying coordinate $x = e^t$. Then

$$\partial_t = x\partial_x, \quad dt = \frac{dx}{x}.$$

With respect to $dt' = dx'/x'$, the inverse of P has kernel

$$K_{P^{-1}}(x, x') = \frac{1}{2\epsilon} e^{-\epsilon|\log(x/x')|} = \frac{1}{2\epsilon} \begin{cases} (x'/x)^\epsilon, & x' \leq x, \\ (x/x')^\epsilon, & x \leq x'. \end{cases}$$

On X_b^2 this kernel is conormal to Δ_b , smooth across the front face away from Δ_b , and vanishes to order ϵ at lb and rb. This model explains why the b -calculus resolves the simultaneous limit $x, x' \rightarrow 0$ while retaining the translation-invariant variable $\log(x/x')$.

5.2 General and iterated blow-ups

The construction is not limited to a corner of a double space. A p -submanifold S of a manifold with corners Z is a submanifold that, in suitable product coordinates

$$(x_1, \dots, x_k, y_1, \dots, y_\ell),$$

is given by the vanishing of a subset of the x - and y -coordinates. Its inward spherical normal bundle is therefore well defined, and the real blow-up

$$[Z; S] = (Z \setminus S) \sqcup S^+ NS$$

has a canonical smooth structure.

In the model $Z = [0, \infty)^2$, $S = \{(0, 0)\}$, the two projective charts are

$$U_x : (x, s = x'/x), \quad U_{x'} : (s' = x/x', x').$$

On their overlap,

$$s' = s^{-1}, \quad x' = sx.$$

These transition functions are smooth where $s, s' > 0$, and the front face is $x = 0$ in the first chart and $x' = 0$ in the second.

$$\begin{array}{ccc} & \xrightarrow{x' = sx, s' = s^{-1}} & \\ U_x = \{(x, s)\} & & U_{x'} = \{(s', x')\} \\ & \xleftarrow{x = s'x', s = (s')^{-1}} & \end{array}$$

Proposition 5.2 (Coordinate invariance). *The blow-up $[Z; S]$, its front face, and its blow-down map are independent up to canonical diffeomorphism of the product coordinates used in the construction.*

Proof. A product-coordinate change preserving S has, in the normal variables, the form

$$\tilde{z} = A(w)z + O(|z|^2),$$

where $A(w)$ is invertible and preserves the inward normal cone. Writing $z = r\omega$ gives

$$\tilde{r} = r |A(w)\omega + O(r)|, \quad \tilde{\omega} = \frac{A(w)\omega + O(r)}{|A(w)\omega + O(r)|}.$$

Both expressions are smooth down to $r = 0$; the second restricts there to the induced map on the spherical normal bundle. Hence every allowed coordinate change lifts smoothly, and the lifted changes satisfy the same cocycle identities. $\square$

Iterated blow-ups are used for X_b^3 and $X_{b,h}^2$. If two centers are disjoint or meet transversely, their blow-ups commute. If one center is contained in another, blowing up the smaller center first makes the lift of the larger one a p-submanifold. These facts justify the rule used below: blow up nested centers in increasing order of dimension and then blow up the transverse lifts.

Discussion. The order is geometric, not cosmetic. Blowing up a larger center first can destroy the product structure of a smaller center and make its lift ill-defined. The triple-space construction specifies the order precisely so that all three lifted projections become b-fibrations.

6 The b-tangent and b-cotangent bundles

Let $\mathcal{V}_b(X)$ be the Lie algebra of smooth vector fields tangent to every boundary face. In local coordinates

$$(x_1, \dots, x_k, y_1, \dots, y_{n-k}) \in [0, \infty)^k \times \mathbb{R}^{n-k},$$

it is the locally free $C^\infty(X)$ -module generated by

$$x_1 \partial_{x_1}, \dots, x_k \partial_{x_k}, \partial_{y_1}, \dots, \partial_{y_{n-k}}.$$

Consequently there is a vector bundle ${}^bTX \rightarrow X$ with

$$C^\infty(X; {}^bTX) = \mathcal{V}_b(X).$$

The natural anchor ${}^bTX \rightarrow TX$ is an isomorphism in the interior but drops rank at the boundary.

The dual bundle ${}^bT^*X$ has local frame

$$\frac{dx_1}{x_1}, \dots, \frac{dx_k}{x_k}, dy_1, \dots, dy_{n-k}.$$

A positive b -density is locally

$$a(x, y) \left| \frac{dx_1}{x_1} \cdots \frac{dx_k}{x_k} dy_1 \cdots dy_{n-k} \right|, \quad a > 0.$$

Under the cylindrical variables $x_j = e^{t_j}$, this is the ordinary product density $a |dt dy|$.

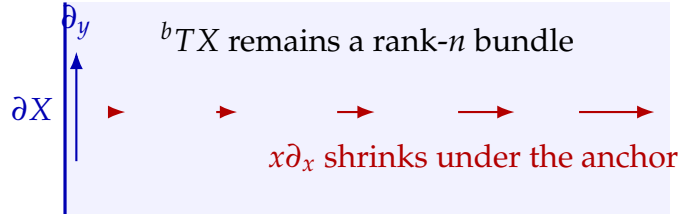


Figure 4: A b-vector field is tangent to the boundary. The normal generator $x\partial_x$ is nonzero in bTX , although its image in TX vanishes at $x = 0$.

7 b-differential operators and their symbols

The algebra $\text{Diff}_b^m(X; E, F)$ consists of differential operators of order at most m generated over $C^\infty(X)$ by $\mathcal{V}_b(X)$. Locally,

$$P = \sum_{|\alpha|+|\beta| \leq m} a_{\alpha\beta}(x, y) (x\partial_x)^\alpha \partial_y^\beta.$$

For manifolds with corners, $(x\partial_x)^\alpha$ denotes $\prod_j (x_j \partial_{x_j})^{\alpha_j}$.

The filtration by order satisfies

$$[\text{Diff}_b^m, \text{Diff}_b^\ell] \subseteq \text{Diff}_b^{m+\ell-1}.$$

It follows that the associated graded algebra is canonically $\text{Sym}({}^bTX)$. Thus the principal symbol is the coordinate-independent homogeneous bundle map

$${}^b\sigma_m(P) : {}^bT^*X \setminus 0 \longrightarrow \text{Hom}(\pi^*E, \pi^*F).$$

In the coordinates above,

$${}^b\sigma_m(P)(x, y; \xi, \eta) = \sum_{|\alpha|+|\beta|=m} a_{\alpha\beta}(x, y) (i\xi)^\alpha (i\eta)^\beta.$$

The symbol is multiplicative:

$${}^b\sigma_{m+\ell}(PQ) = {}^b\sigma_m(P) {}^b\sigma_\ell(Q),$$

and fits into the exact sequence

$$0 \longrightarrow \text{Diff}_b^{m-1}(X; E, F) \longrightarrow \text{Diff}_b^m(X; E, F) \xrightarrow{{}^b\sigma_m} \mathcal{P}^m({}^bT^*X; \text{Hom}(\pi^*E, \pi^*F)) \longrightarrow 0.$$

Here $\mathcal{P}^m$ denotes smooth fibrewise homogeneous polynomials of degree m .

Definition 7.1. An operator $P \in \text{Diff}_b^m(X; E, F)$ is b -elliptic if ${}^b\sigma_m(P)(\zeta)$ is invertible for every nonzero $\zeta \in {}^bT^*X$.

Fix a positive b -density and Hermitian metrics on the bundles. If $V \in \mathcal{V}_b(X)$, integration by parts gives

$$V^* = -V - \text{div}_b(V).$$

Because $\text{div}_b(V) \in C^\infty(X)$, formal adjoints preserve the calculus:

$$P \in \text{Diff}_b^m(X; E, F) \implies P^* \in \text{Diff}_b^m(X; F, E).$$

This also gives

$${}^b\sigma_m(P^*) = {}^b\sigma_m(P)^*.$$

8 Conormal functions and boundary weights

Let X be a compact manifold with corners, with boundary hypersurfaces $H_1, \dots, H_N$ and defining functions $\rho_1, \dots, \rho_N$.

Definition 8.1. The space of bounded conormal functions is

$$\mathcal{A}^0(X) = \{u \in L^\infty(X^\circ) : V_1 \cdots V_k u \in L^\infty(X^\circ) \text{ for all } V_j \in \mathcal{V}_b(X)\}.$$

For a multiweight $\gamma = (\gamma_1, \dots, \gamma_N)$, set

$$\rho^\gamma \mathcal{A}^0(X) = \rho_1^{\gamma_1} \cdots \rho_N^{\gamma_N} \mathcal{A}^0(X).$$

This definition is independent of the chosen defining functions. Indeed, a second defining function has the form $\tilde{\rho}_j = a_j \rho_j$, where a_j is smooth and positive, and multiplication by $a_j^{\gamma_j}$ preserves $\mathcal{A}^0(X)$.

Conormality records uniform control under vector fields tangent to every boundary face. It is weaker than smoothness: powers and logarithms of boundary defining functions are conormal whenever they remain bounded with the prescribed weight.

Definition 8.2. For $\epsilon > 0$, write

$$\mathcal{A}^{0,\epsilon}(X) = \mathcal{C}^\infty(X) + \rho^\epsilon \mathcal{A}^0(X), \quad \rho^\epsilon = \prod_{j=1}^N \rho_j^\epsilon.$$

Thus $u \in \mathcal{A}^{0,\epsilon}(X)$ has a smooth boundary value and a conormal remainder that vanishes at every boundary hypersurface.

The same notation applies to sections of a smooth vector bundle after choosing local trivializations; the resulting space is independent of those trivializations.

9 Conormal distributions and the small b-calculus

Let $Z \subset M$ be an embedded submanifold of codimension k . In local coordinates (x, y) , with $Z = \{x = 0\}$, a classical conormal distribution has the form

$$u(x, y) = (2\pi)^{-k} \int_{\mathbb{R}^k} e^{ix \cdot \xi} a(y, \xi) d\xi,$$

modulo a smooth term, where a is a classical symbol. The symbol order, with the standard dimension shift, defines the conormal order m and the space $I^m(M, Z)$. This description is invariant under coordinate changes.

For vector bundles $E, F \rightarrow X$, let ${}^b\Omega_R$ denote the lifted right b-density bundle on X_b^2 . Set

$$\mathcal{H}_{E,F} = \text{Hom}(E, F) \otimes {}^b\Omega_R.$$

Definition 9.1. The small b-pseudodifferential calculus is

$$\Psi_b^m(X; E, F) = \left\{ A : K_A \in I^m\left(X_b^2, \Delta_b; \mathcal{H}_{E,F}\right), \quad K_A \text{ vanishes to infinite order at } \text{lb} \cup \text{rb} \right\}.$$

The convention relating m to the conormal order depends on the density normalization; throughout these notes, m is the operator order. The lifted diagonal Δ_b meets the front face transversely, so restricting the kernel to the front face gives another conormal distribution.

Every b-differential operator of order m belongs to $\Psi_b^m(X)$: its Schwartz kernel is a finite sum of derivatives of the delta distribution on Δ_b . Operators in the small calculus act continuously

$$\Psi_b^m(X; E, F) : \mathcal{C}^\infty(X; E) \longrightarrow \mathcal{C}^\infty(X; F)$$

and preserve the space $\mathcal{C}^\infty(X)$ of sections vanishing to infinite order at the boundary.

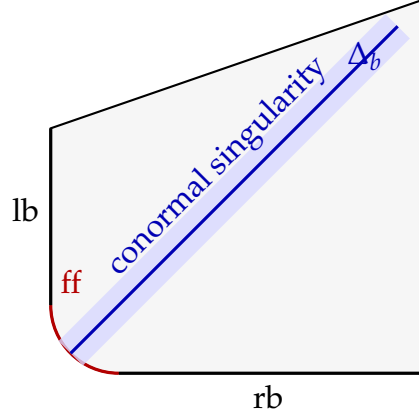


Figure 5: A small-calculus kernel is conormal to the lifted diagonal, extends to the front face, and vanishes rapidly at the two side faces.

The principal symbol is a homogeneous bundle map on ${}^bT^*X \setminus 0$,

$${}^b\sigma_m(A) : \pi^*E \longrightarrow \pi^*F,$$

and fits into the exact sequence

$$0 \longrightarrow \Psi_b^{m-1}(X; E, F) \longrightarrow \Psi_b^m(X; E, F) \xrightarrow{{}^b\sigma_m} S^{[m]}({}^bT^*X; \text{Hom}(E, F)) \longrightarrow 0.$$

Here $S^{[m]} = S^m / S^{m-1}$. An operator is b-elliptic when this symbol is invertible away from the zero section.

10 Composition in the b-calculus

Composition is most transparent on the b-triple space. Start with X^3 and blow up the triple corner and the lifted pairwise corners in increasing order of dimension. The resulting manifold with corners X_b^3 has b-fibrations

$$\pi_{12}, \pi_{23}, \pi_{13} : X_b^3 \longrightarrow X_b^2$$

lifting the three coordinate projections.

Their roles in composition are displayed by

$$\begin{array}{ccccc} & & X_b^3 & & \\ & \swarrow \pi_{12} & \downarrow \pi_{13} & \searrow \pi_{23} & \\ X_b^2 & & X_b^2 & & X_b^2. \end{array}$$

The left and right arrows pull back the two kernels; the middle arrow integrates out the shared variable.

If $A \in \Psi_b^m(X; E, F)$ and $B \in \Psi_b^{m'}(X; G, E)$, then

$$K_{A \circ B} = (\pi_{13})_* (\pi_{12}^* K_A \pi_{23}^* K_B).$$

The lifted conormal sets meet cleanly, and the vanishing at the left and right faces makes the push-forward integrable. The pull-back, product, and push-forward theorems for conormal distributions therefore give

$$\Psi_b^m(X; E, F) \circ \Psi_b^{m'}(X; G, E) \subseteq \Psi_b^{m+m'}(X; G, F).$$

The leading oscillatory amplitudes multiply, so

$${}^b\sigma_{m+m'}(A \circ B) = {}^b\sigma_m(A) {}^b\sigma_{m'}(B).$$

To see the order $m + m'$ directly, work near the lifted diagonal and write the two kernels as oscillatory integrals. Their product has amplitude of order $m + m'$. Stationary phase in the intermediate variable identifies the two covectors and produces the usual symbol product. Away from the lifted diagonal, at least one factor is smooth; the push-forward is then smooth. At every boundary face, the exponent contributed by the product is the sum of the pulled-back exponents minus the fibre-density exponent. The infinite-order side-face vanishing in the small calculus places that sum strictly above the integrability threshold.

The residual part of the full calculus allows polyhomogeneous expansions at the boundary faces. In the positive-weight case used below, products remain integrable and the index sets combine under the same pull-back and push-forward formula. Precise index-set bookkeeping is needed only when a leading exponent reaches an integrability threshold.

11 Parametrices in the small calculus

Let $P \in \Psi_b^m(X; E, F)$ be b-elliptic. Choose $Q_0 \in \Psi_b^{-m}(X; F, E)$ with

$${}^b\sigma_{-m}(Q_0) = {}^b\sigma_m(P)^{-1}.$$

Then

$$PQ_0 = I - R_1, \quad R_1 \in \Psi_b^{-1}(X; F).$$

The symbolic Neumann construction successively removes the homogeneous terms of R_1 . Borel summation produces $Q \in \Psi_b^{-m}(X; F, E)$ such that

$$PQ = I - R_R, \quad QP = I - R_L, \quad R_R, R_L \in \Psi_b^{-\infty}(X).$$

This is a symbolic parametrix, not yet a Fredholm inverse. A residual kernel in the small b-calculus need not be compact on an unweighted b-Sobolev space: after passage to the cylindrical coordinate $t = -\log x$, it may retain translation-invariant behavior as $t \rightarrow \infty$. The normal operator, introduced next, detects precisely this missing boundary condition.

Question 11.1. *Why is a smooth residual kernel not automatically compact?*

Answer. Compactness on a compact manifold follows from smoothing plus Rellich's lemma. The b -metric turns a boundary collar into an infinite cylinder, so translations can escape every compact set. If a residual operator has a nonzero translation-invariant limit $K_\infty(t - t', y, y')$, choose a fixed compactly supported section u on the cylinder and translate it to $u_j(t, y) = u(t - j, y)$. The sequence u_j is bounded and has no convergent subsequence, while Ku_j is asymptotic to the corresponding translate of $K_\infty u$. Hence K is not compact unless its boundary model vanishes or decays. $\square$

12 The normal operator and indicial family

The principal b -symbol controls high-frequency behavior, but it does not control escape along the cylindrical end. The missing invariant is the restriction of the lifted kernel to the front face.

Definition 12.1. For $A \in \Psi_b^m(X; E, F)$, the normal operator $N(A)$ is the translation-invariant operator on the inward normal cylinder of ∂X obtained by restricting the lifted kernel K_A to the front face.

After choosing a boundary defining function x and writing $t = -\log x$, the front face is modeled by $\partial X \times \partial X \times \mathbb{R}_t$. Thus $N(A)$ is a suspended pseudodifferential operator on ∂X . The normal map is multiplicative:

$$N(AB) = N(A)N(B).$$

It follows either from restriction of the composition formula to the front face or from Fourier transformation in t .

For a b -differential operator

$$P = \sum_{j+|\alpha| \leq m} a_{j\alpha}(x, y) (x \partial_x)^j \partial_y^\alpha,$$

freezing the coefficients at $x = 0$ gives

$$N(P) = \sum_{j+|\alpha| \leq m} a_{j\alpha}(0, y) (-\partial_t)^j \partial_y^\alpha.$$

Definition 12.2. The indicial family of P is

$$I(P, \lambda) = x^{-i\lambda} P x^{i\lambda} \Big|_{\partial X} = \sum_{j+|\alpha| \leq m} a_{j\alpha}(0, y) (i\lambda)^j \partial_y^\alpha.$$

For a b -pseudodifferential operator, $I(A, \lambda)$ is the Fourier transform of $N(A)$ in the translation variable, with

$$\widehat{f}(\lambda) = \int_{\mathbb{R}} e^{it\lambda} f(t) dt.$$

Changing x conjugates the indicial family by a smooth nonvanishing factor, so its invertibility is intrinsic. Multiplicativity gives

$$I(AB, \lambda) = I(A, \lambda)I(B, \lambda).$$

The two symbol maps control complementary regimes:

$$\begin{array}{ccc} \Psi_b^m(X) & \xrightarrow{b\sigma_m} & S^{[m]}({}^bT^*X) \\ \downarrow N & & \\ \Psi_{\text{sus}}^m(\partial X) & \xrightarrow{\mathcal{F}_t} & \{I(A, \lambda)\}_{\lambda \in \mathbb{R}}. \end{array}$$

The principal symbol sees large covectors at a fixed point. The indicial family sees bounded frequency while the base point escapes down the cylinder. Full ellipticity requires invertibility in both regimes.

Discussion. This is why the normal operator is often called a second symbol. It is not lower-order information: a smoothing b -operator has zero principal symbol but may have a nonzero normal operator, and that nonzero boundary model can destroy compactness.

For a product b -Dirac operator

$$D = c\left(\frac{dx}{x}\right)(x\partial_x + A_\partial),$$

one has

$$I(D, \lambda) = c\left(\frac{dx}{x}\right)(i\lambda + A_\partial).$$

Consequently D is invertible at the boundary on the unweighted line precisely when the tangential operator A_∂ is invertible.

13 Weighted b -Sobolev spaces

Fix a positive b -density and a positive b -Laplacian. For $s \in \mathbb{R}$, the b -Sobolev space $H_b^s(X; E)$ is defined by b -derivatives for integral s , and by the b -pseudodifferential calculus for arbitrary s . Near the boundary this is the ordinary Sobolev space in the variables $(t, y) = (-\log x, y)$.

For $\delta \in \mathbb{R}$, set

$$x^\delta H_b^s(X; E) = \{x^\delta u : u \in H_b^s(X; E)\}.$$

A b -operator of order m acts continuously as

$$P : x^\delta H_b^s(X; E) \longrightarrow x^\delta H_b^{s-m}(X; F).$$

Conjugation reduces this mapping to the unweighted one:

$$x^{-\delta} P x^\delta : H_b^s(X; E) \longrightarrow H_b^{s-m}(X; F).$$

With the convention $x^{i\lambda}$ used above, the weight x^δ corresponds to the Mellin line

$$\operatorname{Im} \lambda = -\delta.$$

Indeed, write $\lambda = \tau - i\delta$. Then $x^{i\lambda} = x^\delta x^{i\tau}$. Thus conjugating by x^δ moves the real Fourier line for the cylindrical variable to $\operatorname{Im} \lambda = -\delta$; no sign convention is hidden.

Definition 13.1. A b-elliptic operator P is fully elliptic at weight δ if

$$I(P, \lambda) \text{ is invertible whenever } \operatorname{Im} \lambda = -\delta.$$

For several boundary hypersurfaces, impose the corresponding condition at each face with a multiweight. Below, X has one boundary hypersurface.

14 Meromorphic inversion of the indicial family

Suppose P is b-elliptic. Its indicial family is elliptic on the closed manifold ∂X . A parameter-dependent parametrix shows that $I(P, \lambda)$ is invertible for λ in suitable sectors with large modulus. The analytic Fredholm theorem then applies.

Theorem 14.1 (Analytic Fredholm theorem). *Let $T(\lambda)$ be a holomorphic family of Fredholm operators of index zero on a connected domain. If $T(\lambda_0)$ is invertible at one point, then $T(\lambda)^{-1}$ is meromorphic. Every coefficient in the principal part of a Laurent expansion at a pole has finite rank.*

Proof. Fix λ_* . Choose closed complements

$$X = \ker T(\lambda_*) \oplus X_1, \quad Y = \operatorname{ran} T(\lambda_*) \oplus C,$$

where C is finite-dimensional. Since the index is zero, $\dim C = \dim \ker T(\lambda_*)$. Relative to these decompositions, write

$$T(\lambda) = \begin{pmatrix} a(\lambda) & b(\lambda) \\ c(\lambda) & d(\lambda) \end{pmatrix}.$$

The block $d(\lambda_*) : X_1 \rightarrow \operatorname{ran} T(\lambda_*)$ is an isomorphism, so $d(\lambda)^{-1}$ is holomorphic near λ_* . Block Gaussian elimination reduces invertibility of $T(\lambda)$ to that of the finite-dimensional Schur complement

$$S(\lambda) = a(\lambda) - b(\lambda)d(\lambda)^{-1}c(\lambda) : \ker T(\lambda_*) \longrightarrow C.$$

Where S is invertible, the block inverse formula expresses $T(\lambda)^{-1}$ using $d(\lambda)^{-1}$ and $S(\lambda)^{-1}$. Cramer's rule gives

$$S(\lambda)^{-1} = \frac{\operatorname{adj} S(\lambda)}{\det S(\lambda)},$$

so this inverse is meromorphic. Its negative Laurent coefficients have range contained in a finite-dimensional space obtained from C and $\ker T(\lambda_*)$.

It remains to exclude $\det S \equiv 0$ on every overlapping neighborhood. That would make $T(\lambda)$ noninvertible throughout the connected domain, contrary to invertibility at λ_0 . Analytic continuation therefore makes the local meromorphic inverses agree and produces the claimed global meromorphic inverse. $\square$

Applied to $I(P, \lambda)$, this theorem yields a discrete set

$$\text{Spec}_b(P) = \{\lambda \in \mathbb{C} : I(P, \lambda) \text{ is not invertible}\},$$

called the boundary spectrum or set of indicial roots. The inverse $I(P, \lambda)^{-1}$ is meromorphic with finite-rank residues.

A weight is admissible exactly when its Mellin line avoids $\text{Spec}_b(P)$. Moving the line without crossing an indicial root does not change the Fredholm index. When a root is crossed, the index jump is determined by its algebraic multiplicity; the sign depends on the direction in which the weight is moved.

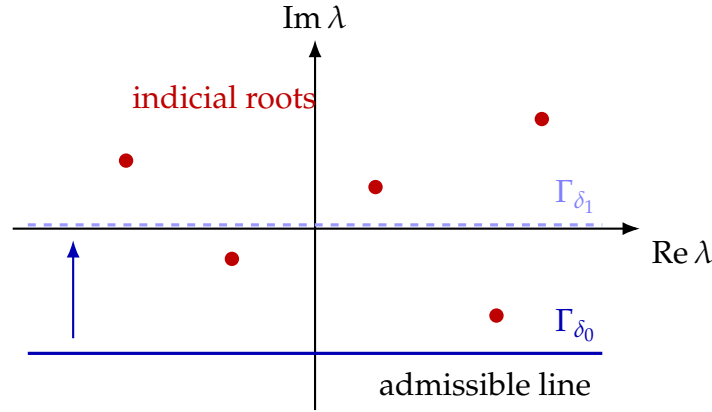


Figure 6: Changing the weight moves the Mellin contour. The index is constant between indicial roots and jumps when the contour crosses one.

15 The full b-calculus and the Fredholm parametrix

The symbolic parametrix constructed in the small calculus has smooth remainders, but their normal operators need not vanish. Full ellipticity allows one to remove this boundary error.

Let $Q_0 \in \Psi_b^{-m}$ be a symbolic parametrix for P :

$$PQ_0 = I - R_0, \quad Q_0P = I - R'_0, \quad R_0, R'_0 \in \Psi_b^{-\infty}.$$

On an admissible Mellin line, define the boundary correction by

$$I(Q_1, \lambda) = I(Q_0, \lambda) + I(P, \lambda)^{-1}I(R_0, \lambda).$$

Then $I(PQ_1, \lambda) = I$. The inverse Mellin transform produces a kernel with polyhomogeneous expansions at the boundary faces. Its exponents are governed by the poles of $I(P, \lambda)^{-1}$. This is the reason the full b-calculus, rather than only the small calculus, is needed.

Here are the details of this correction. Choose a horizontal line $\Gamma_\delta = \{\text{Im } \lambda = -\delta\}$ avoiding the boundary spectrum. With cutoffs $\chi(x)\chi(x')$, the inverse Mellin formula

$$K_C(x, y; x', y') = \frac{1}{2\pi} \int_{\Gamma_\delta} \left(\frac{x}{x'}\right)^{i\lambda} I(P, \lambda)^{-1} I(R_0, \lambda)(y, y') d\lambda$$

defines a residual correction C near the front face. Parameter-elliptic estimates make the integral rapidly convergent in $\text{Re } \lambda$. Applying P , using Mellin multiplicativity, and restricting to the front face gives

$$N(P(Q_0 + C) - I) = 0.$$

If the contour is shifted, the residue theorem adds terms

$$x^{i\lambda_0} (\log x)^j a_{\lambda_0, j}(y, y'),$$

where λ_0 is an indicial root and j is one less than the order of the corresponding pole. These are precisely the polyhomogeneous terms allowed in the full calculus.

The construction is performed on both sides. Any mismatch between the two corrections is residual and has zero normal operator. A Neumann iteration in the residual ideal removes its successive boundary coefficients; Borel summation removes all of them at once. An interior symbolic correction then restores infinite-order vanishing at the lifted diagonal. This yields the two-sided parametrix below.

$$\begin{array}{ccccc} & \text{symbolic inversion} & \text{invert } I(P, \lambda) & & \\ Q_0 & \longrightarrow & R_0 & \longrightarrow & C_{\text{ff}} \\ & & & & \downarrow \text{inverse Mellin} \\ & & \text{remove boundary error} & & \\ Q & \longleftarrow & N(R_1) = 0 & \longleftarrow & Q_0 + C_{\text{ff}} \\ & \text{residual iteration} & & & \end{array}$$

Figure 7: The full parametrix alternates interior symbol inversion with indicial correction at the front face; the remaining residual error is then iterated away.

Iterating at the interior diagonal and at the boundary faces gives $Q \in \Psi_b^{-m}$ in the full calculus with

$$PQ = I - R_R, \quad QP = I - R_L.$$

At an admissible weight, R_R and R_L are residual operators whose normal operators vanish and whose side-face exponents have the decay required by that weight.

Lemma 15.1. *The remainders R_R and R_L act compactly on weighted b -Sobolev spaces.*

Proof. A residual kernel gains arbitrarily many b -derivatives. Vanishing of its normal operator removes the translation-invariant limit on the cylindrical end, while its positive side-face weights give uniform decay there. Cut the cylinder at $t = T$. On the compact part, Rellich's lemma makes the operator compact; on the tail, the operator norm tends to zero as $T \rightarrow \infty$. Hence the full operator is a norm limit of compact operators.

More explicitly, let χ_T equal one on $t \leq T$ and vanish on $t \geq T + 1$. The operator $\chi_T R \chi_T$ factors as

$$H_b^s \xrightarrow{R} H_b^{s+N} \xrightarrow{\chi_T} H_b^s$$

for any $N > 0$; the second arrow is compact by Rellich on the truncated cylinder. The full-calculus estimates and $N(R) = 0$ give

$$\|(1 - \chi_T)R\| + \|R(1 - \chi_T)\| \longrightarrow 0.$$

Therefore

$$R - \chi_T R \chi_T = (1 - \chi_T)R + \chi_T R(1 - \chi_T)$$

tends to zero in operator norm. This proves compactness without assuming that the noncompact b -manifold itself has a compact Sobolev embedding. $\square$

Theorem 15.2 (Fredholm theorem for b -operators). *Let $P \in \Psi_b^m(X; E, F)$ be b -elliptic and fully elliptic at weight δ . Then, for every $s \in \mathbb{R}$,*

$$P : x^\delta H_b^s(X; E) \longrightarrow x^\delta H_b^{s-m}(X; F)$$

is Fredholm.

Proof. The full b -parametrix Q is bounded between the reverse weighted Sobolev spaces, and

$$PQ = I - R_R, \quad QP = I - R_L$$

with compact remainders. Atkinson's theorem therefore implies that P is Fredholm. $\square$

Corollary 15.3 (Elliptic regularity). *If $Pu = f$, where $u \in x^\delta H_b^{s_0}$ and $f \in x^\delta H_b^{s-m}$, then $u \in x^\delta H_b^s$. In particular, null-solutions are smooth in the interior and polyhomogeneous at the boundary, with exponents drawn from the indicial spectrum.*

Proof. Apply the parametrix:

$$u = Qf + R_L u.$$

The first term gains m derivatives, and the residual term gains arbitrarily many. Iteration gives the stated regularity. Mellin inversion and contour shifting yield the boundary expansion; residues at indicial poles produce powers of x and, at higher-order poles, logarithmic terms. $\square$

16 The Dirac case

For

$$D = c\left(\frac{dx}{x}\right)(x\partial_x + A_\partial),$$

the b -principal symbol is Clifford multiplication and is invertible away from the zero section. Thus D is b -elliptic. Its indicial family is invertible on the unweighted line if and only if A_∂ is invertible.

If A_∂ has kernel, one may choose a small nonzero weight δ whose Mellin line avoids the boundary spectrum. The resulting weighted realization is Fredholm, but its index depends on the side from which the zero indicial roots are crossed. The APS projection selects the corresponding spectral subspace; the correction at zero is $\frac{1}{2} \dim \ker A_\partial$ in the index formula.

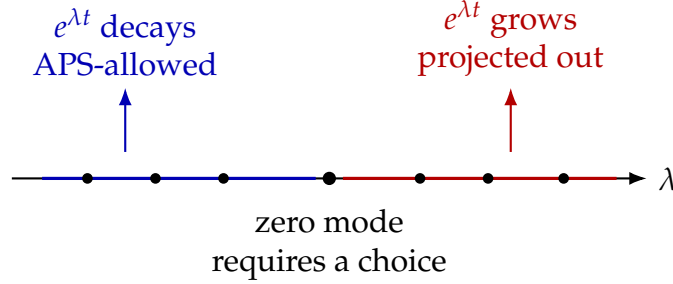


Figure 8: On the outward cylinder, the tangential spectrum splits boundary data into decaying and growing modes. The APS projection makes this split at $\lambda = 0$.

17 The K-theoretic proof architecture

The closed index theorem has two layers: an elliptic symbol defines a K -class, and the analytic index is the unique push-forward of that class normalized by the Bott operator. On a manifold with boundary, the interior symbol alone is insufficient. One must also record an invertible model at the cylindrical end. The b-calculus packages exactly these two pieces [2, 3].

Let $\mathcal{A}_b$ be the operator-norm closure of the stabilized order-zero b-calculus, and let Σ_b be its joint symbol algebra. An element of Σ_b is a compatible pair

$$(a, q) = ({}^b\sigma_0(P), I(P, \lambda)),$$

where a is the b-principal symbol and q is the suspended indicial family. Compatibility means that the principal symbol of q is the restriction of a to the boundary. The full symbol sequence is

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{A}_b \xrightarrow{j} \Sigma_b \longrightarrow 0.$$

Indeed, a b-operator with zero principal symbol and zero normal operator is compact; the parametrix construction proves exactness at the other terms.

If P is fully elliptic, $j(P)$ is invertible. After stabilization it defines a class $[j(P)] \in K_1(\Sigma_b)$. The six-term exact sequence has a connecting map

$$\partial : K_1(\Sigma_b) \longrightarrow K_0(\mathcal{K}) \cong \mathbb{Z}.$$

Lemma 17.1 (The connecting map is the Fredholm index). *With the usual sign convention for ∂ ,*

$$\partial[j(P)] = \text{ind } P.$$

Proof. Lift $j(P)^{-1}$ to a b-parametrix Q . Then

$$I - QP \in \mathcal{K}, \quad I - PQ \in \mathcal{K}.$$

After a compact perturbation, the polar part V of P is a partial isometry. The definition of the connecting map gives

$$\partial[j(P)] = [I - V^*V] - [I - VV^*].$$

The first defect projection is the orthogonal projection onto $\ker P$, and the second is the projection onto $\ker P^*$. Under $K_0(\mathcal{K}) \cong \mathbb{Z}$, their difference is $\dim \ker P - \dim \ker P^* = \text{ind } P$. $\square$

For the product Dirac operator, the unbounded joint elliptic data are

$$\left({}^b\sigma_1(D), I(D, \lambda) = c\left(\frac{dx}{x}\right) (i\lambda + A) \right).$$

Composing on the right with the inverse of a fixed positive elliptic family of order one normalizes this to an order-zero element of Σ_b without changing invertibility or the index class. Equivalently, one may use the bounded transform of D , whose indicial family includes the corresponding factor of order minus one. If A is invertible, this pair is fully elliptic. If A has kernel, replace it in the indicial family by

$$A_\varepsilon = A + \varepsilon \Pi_0, \quad \varepsilon > 0,$$

where Π_0 projects onto $\ker A$. This smoothing perturbation is finite rank, and the perturbed family is invertible. It chooses the same side of the zero modes as the APS projection $\Pi_{\geq 0}(A)$. Thus the APS condition is not extra data unrelated to the symbol: it is the boundary trivialization needed to make the joint symbol a K -class.

Denote the resulting stable class by

$$[\sigma_{\text{APS}}(D^+)] \in K_{\text{APS}}^0({}^bT^*X).$$

This notation means stable homotopy classes of b-symbols equipped with their invertible boundary trivializations. A neat embedding

$$i : (X, \partial X) \hookrightarrow (\mathbb{R}_+^N, \mathbb{R}^{N-1})$$

defines a Thom push-forward of these relative classes. Normal deformation and collar doubling identify the target with the Bott group

$$K_c^0(\mathbb{R}^{2N}) \cong \mathbb{Z}.$$

Call the resulting integer $\text{t-ind}_{\text{APS}}$.

Theorem 17.2 (K-theoretic APS index theorem). *For a product-type graded Dirac operator,*

$$\text{ind}(D_{\text{APS}}^+) = \text{t-ind}_{\text{APS}}[\sigma_{\text{APS}}(D^+)].$$

The theorem asserts that the following diagram commutes; the left vertical map is the connecting-map index of the joint symbol.

$$\begin{array}{ccc}
K_{\text{APS}}^0({}^bT^*X) & \xrightarrow{i_!} & K_c^0(\mathbb{R}^{2N}) \\
\text{a-ind}_{\text{APS}} \downarrow & & \downarrow \text{Bott} \\
\mathbb{Z} & \xlongequal{\quad\quad\quad} & \mathbb{Z}
\end{array}$$

Proof. The proof follows the closed push-forward argument. In a tubular neighborhood, the Thom class is represented analytically by the fibrewise Bott operator

$$B_v = \sum_j c(e_j) \partial_{v_j} + \widehat{c}(v),$$

on $\Lambda^\bullet \nu_{\mathbb{C}}^*$, where $c(v) = \varepsilon(v^b) - \iota_v$ and $\widehat{c}(v) = \varepsilon(v^b) + \iota_v$. Its even kernel is the Gaussian line and its odd kernel is zero. Taking the graded product with D_{APS}^+ therefore preserves the index. On the boundary collar the normal Bott factor is product type, so it carries the chosen indicial trivialization to the pushed-forward operator without changing which zero modes are selected. Excision removes the complement of the tubular neighborhood.

After the neat embedding and collar doubling, the class is an integer multiple of the Bott generator. The Bott operator has index one, so both the analytic index, computed by the connecting map above, and the topological push-forward equal that integer. This proves the formula. $\square$

This theorem proves the equality of two *integer* indices. Ordinary K -theory does not by itself determine the real number

$$\bar{\eta}(A) = \frac{\eta(A) + \dim \ker A}{2}.$$

That number is a secondary, differential refinement of the boundary K -class: it can vary continuously with the metric and connection even when the underlying K -class is fixed. The heat-kernel and b-trace argument below supplies its canonical real lift and proves the numerical formula

$$\text{ind}(D_{\text{APS}}^+) = \int_X \left[\widehat{A}(TX) \text{ch}(E/S) \right]_{[n]} - \bar{\eta}(A).$$

Equivalently,

$$\bar{\eta}(A) \equiv \int_X \left[\widehat{A}(TX) \text{ch}(E/S) \right]_{[n]} \pmod{\mathbb{Z}}.$$

Thus the exponentiated eta invariant is the boundary transgression of the local index form. The K -theoretic proof explains integrality and homotopy invariance, while the heat proof identifies the precise spectral correction.

Question 17.3. *Why can $\bar{\eta}(A)$ vary while the K -theoretic APS index is constant?*

Answer. As the geometric data vary without a zero eigenvalue crossing, the local integral and $\bar{\eta}(A)$ vary by the same amount, so their difference remains the fixed integer index. At a crossing, the spectral projection changes by a finite-dimensional space; spectral flow changes the K -theoretic boundary trivialization and accounts for the integer jump. Ordinary K -theory records that jump, whereas the eta invariant records the continuous real-valued refinement between jumps. $\square$

18 The b-trace defect

Let A be a residual b-operator whose lifted kernel has a full expansion at the front face and sufficient decay at the side faces. Its b-trace is the finite part of the diagonal integral,

$${}^b\mathrm{Tr}(A) = \mathrm{FP}_{z=0} \int_X x^z \mathrm{tr}_E K_A(p, p) d\mu_b(p).$$

The ordinary trace property fails because the finite part retains a boundary contribution.

Use the Fourier convention fixed for the indicial family:

$$\widehat{f}(\lambda) = \int_{\mathbb{R}} e^{it\lambda} f(t) dt, \quad f(t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-it\lambda} \widehat{f}(\lambda) d\lambda.$$

Theorem 18.1 (b-trace defect formula). *If the products below are defined and residual, then*

$${}^b\mathrm{Tr}([A, B]) = \frac{i}{2\pi} \int_{\mathbb{R}} \mathrm{Tr}_{\partial X} (\partial_\lambda I(A, \lambda) I(B, \lambda)) d\lambda.$$

For graded bundles, the same formula holds with the supercommutator and supertrace.

Proof. For $\mathrm{Re} z > 0$, cyclicity of the ordinary trace gives

$$\mathrm{Tr}(x^z[A, B]) = \mathrm{Tr}([x^z, A]B).$$

Set

$$C_z = z^{-1} x^{-z} [x^z, A].$$

This family extends to $z = 0$, and its indicial family there is

$$I(C_0, \lambda) = i \partial_\lambda I(A, \lambda).$$

The trace of $x^z C_z B$ has a simple pole at zero. Its residue is the Fourier integral of $I(C_0, \lambda) I(B, \lambda)$, divided by 2π . Since $[x^z, A]B = z x^z C_z B$, that residue is the constant term of the original trace. Finally $i/(2\pi) = -1/(2\pi i)$. $\square$

This identity is independent of the defining function: changing x changes both the finite part and the indicial family by compensating terms.

19 Dirac operators of product type

Let X be an even-dimensional compact Riemannian manifold with boundary. Assume that the metric and Clifford data are products near the boundary. Let

$$E = E^+ \oplus E^-$$

be a graded Hermitian Clifford module with compatible connection. Its Dirac operator is odd,

$$D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix},$$

and near ∂X it has the form

$$D = c(du)(\partial_u + A),$$

where u is the inward normal coordinate and A is the self-adjoint tangential Dirac operator.

Extend the collar coordinate u to negative values on the attached outward cylinder, put $t = -u \geq 0$, and set $x = e^{-t} = e^u$. Then $du = dx/x$ and $\partial_u = x\partial_x$, so the product formula becomes

$$D = c\left(\frac{dx}{x}\right)(x\partial_x + A),$$

with the same A as above. This convention fixes the sign of the APS correction to be $-(\eta_A(0) + \dim \ker A)/2$.

The Weitzenböck formula is

$$D^2 = \nabla^* \nabla + \mathcal{R},$$

where $\mathcal{R}$ is the curvature endomorphism. For the spin Dirac operator twisted by W ,

$$\mathcal{R} = \frac{1}{4} \text{scal} + \frac{1}{2} \sum_{j,k} c(e_j)c(e_k)F^W(e_j, e_k).$$

20 The b-heat kernel

The heat kernel is constructed on the parabolic b-heat space

$$X_{b,h}^2 = \left[X_b^2 \times [0, \infty)_t; \Delta_b \times \{0\} \right]_{\text{par}},$$

where the final blow-up assigns weight one to spatial variables and weight two to t . The new temporal front face resolves the singularity of the kernel as $t \downarrow 0$.

At the temporal front face, the model equation is the Euclidean heat equation on each b-tangent fibre. At the boundary front face, the model is the heat equation for the normal operator on the cylinder. Full ellipticity makes the large-cylinder model invertible in the required weighted space.

Proposition 20.1 (Compatibility of the heat models). *The Euclidean model at the temporal front face and the cylindrical normal model at the boundary front face have the same restriction to their intersection. They therefore determine a single leading kernel on $X_{b,h}^2$.*

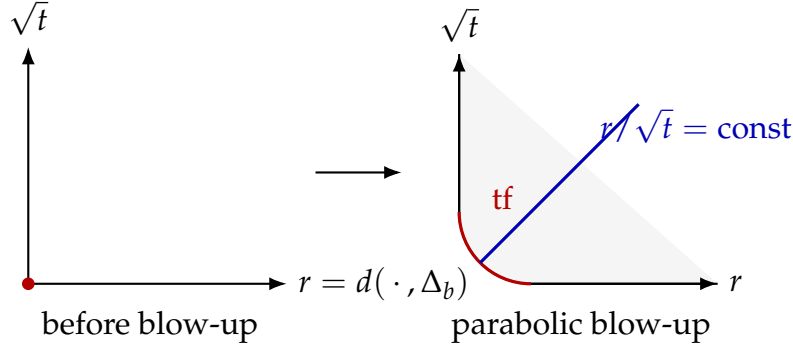


Figure 9: The temporal diagonal is parabolically blown up. The new face keeps the heat variable $r/\sqrt{t}$ finite and carries the Gaussian model.

Proof. Near a boundary point use b-normal coordinates

$$(t, y) = (-\log x, y).$$

After parabolic rescaling about the diagonal, the coefficients of D^2 freeze at the base point. The temporal model is therefore the Gaussian heat kernel of the constant-coefficient operator on bT_pX . If the base point tends to the boundary, the same frozen operator splits into the normal line in t and the tangential operator on T_yY . This is exactly the small-time restriction of the translation-invariant normal heat kernel. Thus both models restrict to

$$(4\pi)^{-n/2} e^{-|V|^2/4}$$

times parallel transport at the corner of the two faces. $\square$

The compatibility permits one to extend the two prescribed normal operators to a global initial parametrix. Successive transport equations remove the Taylor coefficients of the error at the temporal face; successive normal-operator inversions remove its coefficients at the boundary face. Because the corrections agree at the corner, alternating between the two faces improves the error without spoiling work already done at the other face. Borel summation produces a kernel whose heat error vanishes to infinite order at every temporal boundary.

Starting with the Gaussian model, solve the transport equations successively along the temporal front face. Borel summation gives a parametrix $H_0(t)$ satisfying

$$(\partial_t + D^2)H_0(t) = R(t),$$

where R vanishes to infinite order at $t = 0$. The Volterra series

$$H = H_0 - H_0 * R + H_0 * R * R - \dots$$

converges for small t , and the semigroup property extends it to all positive times. Here

$$(K_1 * K_2)(t) = \int_0^t K_1(t-s)K_2(s) ds.$$

Thus $H(t) = e^{-tD^2}$.

The convergence uses the ordered time simplex, a point that is easy to lose in a formal Neumann-series argument. If $\|R(t)\| \leq C$ for $0 \leq t \leq t_0$, then

$$R^{*k}(t) = \int_{\substack{s_j \geq 0 \\ s_1 + \dots + s_k = t}} R(s_1) \cdots R(s_k) ds_1 \cdots ds_{k-1}$$

and the simplex has volume $t^{k-1}/(k-1)!$. Hence

$$\|R^{*k}(t)\| \leq \frac{C^k t^{k-1}}{(k-1)!}.$$

The factorial proves absolute convergence. The same estimate after applying b -derivatives proves convergence in the residual heat calculus. Since R vanishes to infinite order at $t = 0$, so does every correction term, and the initial value remains the identity.

The construction also proves Duhamel's formula. If L_r is a smooth family of nonnegative self-adjoint elliptic operators, then

$$\frac{d}{dr} e^{-tL_r} = - \int_0^t e^{-(t-s)L_r} \dot{L}_r e^{-sL_r} ds.$$

Both sides solve the same inhomogeneous heat equation with zero initial value, which proves the identity by uniqueness.

21 Small-time supertrace

For an ordinary Laplace-type operator, the diagonal heat kernel has an expansion

$$K(t; p, p) \sim (4\pi t)^{-n/2} \sum_{j=0}^{\infty} t^j a_j(p).$$

For a Dirac operator, Clifford supertrace cancels all terms below total degree n . Getzler rescaling identifies the surviving term.

Theorem 21.1 (Local index theorem). *For a graded Dirac operator,*

$$\lim_{t \downarrow 0} \text{Str} K(t; p, p) d \text{vol}(p) = \left[\hat{A}(TX, \nabla^{TX}) \text{ch}(E/S, \nabla^{E/S}) \right]_{[n]}.$$

For the spin Dirac operator twisted by W , the second factor is $\text{ch}(W, \nabla^W)$.

On a manifold with cylindrical end, diagonal integration is interpreted as a b -integral. The heat-space expansion shows

$$\lim_{t \downarrow 0} {}^b \text{Str}(e^{-tD^2}) = \int_X \left[\hat{A}(TX) \text{ch}(E/S) \right]_{[n]}.$$

Because the data are of product type, no additional local boundary density appears.

22 The eta invariant

For $\operatorname{Re} s$ large, define

$$\eta_A(s) = \sum_{\lambda \in \operatorname{Spec}(A) \setminus \{0\}} \operatorname{sign}(\lambda) |\lambda|^{-s}.$$

The Mellin representation is

$$\eta_A(s) = \frac{1}{\Gamma((s+1)/2)} \int_0^\infty t^{(s-1)/2} \operatorname{Tr}(Ae^{-tA^2}) dt.$$

The large-time integral is entire after omission of the zero eigenspace. The heat expansion gives a meromorphic continuation of the small-time integral.

Theorem 22.1. *The eta function of a self-adjoint elliptic differential operator of positive order on a closed manifold is regular at $s = 0$.*

Proof. Split the Mellin integral at $t = 1$. On the orthogonal complement of the kernel, the large-time integrand is exponentially decreasing, so that part is entire. For an operator of order r on a d -manifold, the pseudodifferential heat calculus gives a polyhomogeneous expansion of $\operatorname{Tr}(Ae^{-tA^2})$ as $t \downarrow 0$. Termwise Mellin integration gives the meromorphic continuation.

The possible residue at zero is a constant multiple of the noncommutative residue of the order-zero operator

$$\operatorname{sgn}(A) = A|A|^{-1}$$

on $\ker A^\perp$. Modulo a finite-rank operator, $\operatorname{sgn}(A) = 2\Pi_+ - I$, where Π_+ is the positive spectral projection. The noncommutative residue vanishes on the identity, on finite-rank operators, and on pseudodifferential projections. The possible residue is therefore zero. For a Dirac-type operator the same cancellation is visible directly in the parity of its local Clifford heat coefficients. $\square$

For a Dirac-type operator, regularity is also visible from the parity of the Clifford heat coefficients. Write

$$\eta(A) = \eta_A(0), \quad h(A) = \dim \ker A.$$

23 Heat proof of the APS index theorem

Assume first that A is invertible. Set

$$F(t) = {}^b \operatorname{Str}(e^{-tD^2}).$$

Since D commutes with e^{-tD^2} ,

$$F'(t) = -\frac{1}{2} {}^b \operatorname{Str} \left([D, De^{-tD^2}]_s \right).$$

Apply the graded b-trace defect formula and use

$$I(D, \lambda) = c\left(\frac{dx}{x}\right) (i\lambda + A).$$

Moreover,

$$I(De^{-tD^2}, \lambda) = c\left(\frac{dx}{x}\right) (i\lambda + A)e^{-t(\lambda^2 + A^2)}, \quad \partial_\lambda I(D, \lambda) = i c\left(\frac{dx}{x}\right).$$

The Clifford relation $c(dx/x)^2 = -I$, together with the graded matrix trace, leaves the term containing A ; the term odd in λ integrates to zero. Finally,

$$\int_{\mathbb{R}} e^{-t\lambda^2} d\lambda = \sqrt{\frac{\pi}{t}}.$$

Substitution into the defect formula, including the factor $1/2$ above, gives

$$F'(t) = -\frac{1}{2\sqrt{\pi t}} \text{Tr}(Ae^{-tA^2}).$$

This is the only place where the b-supertrace varies with t .

The full heat kernel and the spectral gap at the boundary imply

$$\lim_{t \rightarrow \infty} F(t) = \text{ind}(D_{\text{APS}}^+).$$

The local index theorem gives

$$\lim_{t \downarrow 0} F(t) = \int_X \left[\hat{A}(TX) \text{ch}(E/S) \right]_{[n]}.$$

Integrating the trace-defect identity and using the Mellin formula at $s = 0$ yields

$$\text{ind}(D_{\text{APS}}^+) = \int_X \left[\hat{A}(TX) \text{ch}(E/S) \right]_{[n]} - \frac{1}{2} \eta(A).$$

Indeed,

$$F(\infty) - F(0) = -\frac{1}{2\sqrt{\pi}} \int_0^\infty t^{-1/2} \text{Tr}(Ae^{-tA^2}) dt = -\frac{1}{2} \eta_A(0).$$

At the lower endpoint, this means: first insert $t^{s/2}$ for $\text{Re } s$ large, use the Mellin identity, and then continue to $s = 0$.

If A has kernel, perturb it by a finite-rank operator on $\ker A$, or equivalently use the APS spectral projection and pass to the limit. The zero modes contribute $h(A)/2$.

To determine the contribution and its sign, put $A_\varepsilon = A + \varepsilon \Pi_0$, where Π_0 projects onto $\ker A$ and $\varepsilon > 0$. Then

$$\eta(A_\varepsilon) = \eta(A) + h(A).$$

The decaying-mode condition on the perturbed cylinder converges to $\Pi_{\geq 0}(A)(u|_{\partial X}) = 0$. The Fredholm index is constant for $\varepsilon > 0$, so passage to the limit gives $-(\eta(A) + h(A))/2$.

Theorem 23.1 (Atiyah–Patodi–Singer). *Let D^+ be a product-type graded Dirac operator on an even-dimensional compact manifold with boundary. With APS boundary condition*

$$\Pi_{\geq 0}(A)(u|_{\partial X}) = 0,$$

one has

$$\boxed{\operatorname{ind}(D_{\text{APS}}^+) = \int_X \left[\widehat{A}(TX) \operatorname{ch}(E/S) \right]_{[n]} - \frac{\eta_A(0) + \dim \ker A}{2}.$$

The interior term is local. The eta invariant and the kernel dimension are the global spectral correction forced by the cylindrical end.

Question 23.2. *Why can the boundary correction not be another local curvature integral?*

Answer. The derivative $F'(t)$ is the full spectral trace $\operatorname{Tr}(Ae^{-tA^2})$, not the integral of one fixed heat coefficient. Integrating it from zero to infinity uses every eigenvalue of A . Small-time asymptotics control the meromorphic continuation, but the value $\eta_A(0)$ also contains large-time information and therefore depends on the global spectral asymmetry. This is exactly the information lost when a local index density is integrated on a manifold with boundary. $\square$

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